Point A and B is still something of a mystery, as is what it actually *means* that particles are *measured* as small localized things but still manage to obey the *mathematics* of waves that are spread out through space.

The one thing everyone agrees on is what the data say, and those data make it very clear that tunneling through seemingly impassable barriers is something that particles are very happy to do on the regular. Wherever a particle really goes in the interim, it's clear that a wall can't stop it. This sort of escape artistry is such normal behavior for particles that people who design things like cell phones and microprocessors have to take into account the fact that every once in a while an otherwise well-behaved electron will suddenly materialize on the wrong side of a chip. Some technologies, including flash memory, occasionally also use this to their advantage. Scanning tunneling microscopes use the expectation of tunneling almost like a valve to drip electrons slowly onto a surface and get images of individual atoms.

Letting electrons sneak across short gaps or squeeze through insulating barriers is a nice party trick, but it gets significantly more ominous when you realize that quantum tunneling can be performed not only by particles, but also by fields. Fields like the Higgs, separated from that big true-vacuum valley by a potential barrier it can tunnel right through. The only thing standing between our nice hospitable universe and ultimate cosmic disaster suddenly looks a lot less solid.

The (somewhat) good news is that even something as weird as quantum tunneling does actually follow certain rules, at least when it comes to the expected rate of its occurrence. The probability for a tunneling event is based on the physical characteristics of the system, which means that how likely it is to happen over a set period of time can be very well known. It isn't a total free-for-all. As hard as quantum mechanics may be to fully understand or interpret, it is at least calculable.

But those "rules" that we calculate don't come in any form more reassuring than probabilities. We can't confidently say that the Higgs field *won't* tunnel across the barrier and create a bubble of quantum death right next to you in the next 30 seconds, setting off a process of unthinkable destruction that will tear through space for all eternity. What we can say is: that scenario is extremely unlikely. (At least, the "in the next 30 seconds" part is. If our vacuum really is metastable, strictly speaking, the bubble has to show up eventually.)